

Theorem 3 (Polished): Unidentifiability of Noise vs. Learnable Difficulty

Core claim: From observable data $(x, y, \{f_m(x)\})$ alone, label noise and intrinsic sample difficulty are observationally equivalent. SCX’s assumptions (A1)-(A6) are not arbitrary technical conditions but the minimal sufficient structure needed to break this equivalence.

Revision: 2026-06-27 — Generalized to K -class, arbitrary state count; added minimal-counterexample proof; explicit assumption-corollary mapping; positioned within the measurement-error / label-noise / partial-identification literature.

1 Theorem Statement

1.1 Setup

Let the following be defined as in Theorem 1 (see notation document for full cross-reference):

Symbol	Meaning
$\mathcal{X}$	Input space
$\mathcal{Y}$	Label space, $\ \mathcal{Y}\ = K$
$Y^* \in \mathcal{Y}$	True label (latent , unobservable)
$Y \in \mathcal{Y}$	Observed label

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

Symbol	Meaning
$\{f_m\}_{m=1}^M$	M expert models
$\mathcal{D}_{\text{obs}}$	Observable data distribution, $(X, Y, \{f_m(X)\}) \sim \mathcal{D}_{\text{obs}}$
$\mathcal{S}$	State space
$\eta(x) = \mathbb{P}(Y \neq Y^* \mid X = x)$	Input-dependent noise rate

What is observable: The researcher sees only the joint distribution of $(x_i, y_i, f_1(x_i), \dots, f_M(x_i))$. The true label y^* , the state $s(x)$, and the noise indicator are **all latent**.

1.2 Formal Statement

Theorem 3 (Noise-Difficulty Unidentifiability). For any $K \geq 2$ classes, any $M \geq 1$ experts, and any finite state space $\mathcal{S}$, there exist two distinct data-generating processes $\mathcal{P}_1$ and $\mathcal{P}_2$ such that:

- (i) Under $\mathcal{P}_1$, some observed label errors are caused by **label noise** ($y \neq y^*$).
- (ii) Under $\mathcal{P}_2$, all observed labels equal the true labels ($y = y^*$ for all samples), but some samples have **intrinsic difficulty**: experts systematically err because of sample ambiguity rather than label corruption.
- (iii) Observational equivalence:

$$\mathcal{P}_1(x, y, \{f_m(x)\}) = \mathcal{P}_2(x, y, \{f_m(x)\}), \quad \forall (x, y, \{f_m\})$$

- (iv) Therefore, no algorithm $\mathcal{A} : (\mathcal{X} \times \mathcal{Y} \times \mathcal{Y}^M)^n \rightarrow \{0, 1\}^n$ can correctly identify noise in both worlds simultaneously: there exists n such that $\mathcal{A}$ has

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

error rate $\geq 1/2$ in at least one world.

Interpretation: “This sample’s label is wrong” and “This sample is too hard for the experts” are observationally equivalent propositions. SCX’s assumption set (A1)-(A6) is precisely what is needed to break this equivalence.

2 Proof: Generalized Construction

2.1 Design Principles

The construction is designed to be **minimal**: it uses the fewest moving parts necessary to establish unidentifiability.

Minimality properties: - **2 states** suffice (one ambiguous, one clean): $\mathcal{S} = \{s_1, s_2\}$ - **Binary classification suffices** ($K = 2$): the K -class case is a corollary - **1 expert suffices** ($M = 1$): the $M > 1$ case follows identically - **No additional structure**: the construction does not rely on violating any assumption – it merely shows that within the space of all possible data-generating processes, indistinguishable pairs exist

2.2 World A: Noise-Driven

In $\mathcal{P}_{\text{noise}}$:

State s_1 (“clean and easy”): - $\mathbb{P}(X \in s_1) = \rho$, with $\rho \in (0, 1)$ - Constant true label: for all $x \in s_1$, $y^* = 0$ - Noise mechanism: labels flipped with

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

probability $\eta \in (0, 1/2)$

$$\mathbb{P}(y \neq y^* \mid x \in s_1) = \eta$$

- Expert accuracy on clean samples: $\mathbb{P}(f_m(x) = y^* \mid x \in s_1, \text{clean}) = 1 - \varepsilon_1$,
 $\varepsilon_1 \in (0, 1/2)$

State s_2 (“intrinsically hard”): - $\mathbb{P}(X \in s_2) = 1 - \rho$ - Constant true label:
for all $x \in s_2$, $y^* = 0$ - No noise: $\mathbb{P}(y \neq y^* \mid x \in s_2) = 0$ - Lower expert
accuracy: $\mathbb{P}(f_m(x) = y^* \mid x \in s_2) = 1 - \varepsilon_2$, $\varepsilon_1 < \varepsilon_2 \leq 1/2$

2.3 World B: Difficulty-Driven

In $\mathcal{P}_{\text{hard}}$:

State s_1 (“ambiguous”, noise-free): - $\mathbb{P}(X \in s_1) = \rho$ (same as World A) -
All labels are true: $y = y^*$ for every sample - But the true label itself is
ambiguous:

$$\mathbb{P}(y^* = 0 \mid x \in s_1) = 1 - \eta, \quad \mathbb{P}(y^* = 1 \mid x \in s_1) = \eta$$

where η matches World A’s noise rate - Expert behavior (class-conditional):
- When $y^* = 0$: $\mathbb{P}(f_m = 0 \mid s_1, y^* = 0) = 1 - \varepsilon_1$ - When $y^* = 1$: $\mathbb{P}(f_m = 0 \mid s_1, y^* = 1) = 1 - \varepsilon_1$ (biased toward class 0) - Hence: accuracy is $1 - \varepsilon_1$ when
 $y^* = 0$, but only ε_1 when $y^* = 1$

State s_2 (“hard”, same as World A): - $\mathbb{P}(X \in s_2) = 1 - \rho$, $y = y^* = 0$ for
all samples - Expert accuracy: $\mathbb{P}(f_m(x) = y^* \mid x \in s_2) = 1 - \varepsilon_2$

2.4 Observational Equivalence Verification

Decompose the joint distribution:

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

$$\mathcal{P}(x, y, f_1, \dots, f_M) = \mathcal{P}(x) \cdot \mathcal{P}(y \mid x) \cdot \prod_{m=1}^M \mathcal{P}(f_m \mid x)$$

Step 1: $\mathcal{P}(x)$. $\mathbb{P}(X \in s_1) = \rho$ in both worlds. Identical.

Step 2: $\mathcal{P}(y \mid x)$.

In $\mathcal{P}_{\text{noise}}$ (state s_1):

$$\begin{aligned}\mathcal{P}_{\text{noise}}(y = 0 \mid s_1) &= (1 - \eta) \cdot 1 + \eta \cdot 0 = 1 - \eta \\ \mathcal{P}_{\text{noise}}(y = 1 \mid s_1) &= \eta\end{aligned}$$

In $\mathcal{P}_{\text{hard}}$ (state s_1):

$$\begin{aligned}\mathcal{P}_{\text{hard}}(y = 0 \mid s_1) &= \mathbb{P}(y^* = 0 \mid s_1) = 1 - \eta \\ \mathcal{P}_{\text{hard}}(y = 1 \mid s_1) &= \eta\end{aligned}$$

State s_2 : $\mathcal{P}(y = 0 \mid s_2) = 1$, $\mathcal{P}(y = 1 \mid s_2) = 0$ in both worlds. Identical. ✓

Step 3: $\mathcal{P}(f_m \mid x)$.

In $\mathcal{P}_{\text{noise}}$ (state s_1):

$$\mathcal{P}_{\text{noise}}(f_m = 0 \mid s_1) = 1 - \varepsilon_1, \quad \mathcal{P}_{\text{noise}}(f_m = 1 \mid s_1) = \varepsilon_1$$

In $\mathcal{P}_{\text{hard}}$ (state s_1):

$$\begin{aligned}\mathcal{P}_{\text{hard}}(f_m = 0 \mid s_1) &= (1 - \eta)(1 - \varepsilon_1) + \eta(1 - \varepsilon_1) = 1 - \varepsilon_1 \\ \mathcal{P}_{\text{hard}}(f_m = 1 \mid s_1) &= (1 - \eta)\varepsilon_1 + \eta\varepsilon_1 = \varepsilon_1\end{aligned}$$

State s_2 : $\mathcal{P}(f_m = 0 \mid s_2) = 1 - \varepsilon_2$, $\mathcal{P}(f_m = 1 \mid s_2) = \varepsilon_2$ in both worlds. Identical. ✓

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

Step 4: Joint identity. All factors match; therefore:

$$\mathcal{P}_{\text{noise}}(x, y, \{f_m\}) = \mathcal{P}_{\text{hard}}(x, y, \{f_m\}), \quad \forall (x, y, \{f_m\})$$

Step 5: Algorithmic indistinguishability. Any algorithm $\mathcal{A}$ receiving n i.i.d. samples from either distribution sees identical data. Therefore its output distribution is identical:

$$\mathcal{L}_{\mathcal{P}_{\text{noise}}}(\mathcal{A}(D_n)) = \mathcal{L}_{\mathcal{P}_{\text{hard}}}(\mathcal{A}(D_n))$$

But the true “noise” sets differ: - In $\mathcal{P}_{\text{noise}}$: $\mathcal{Z}_{\text{noise}}^* = \{i : x_i \in s_1, y_i = 1\}$ (these are flipped) - In $\mathcal{P}_{\text{hard}}$: $\mathcal{Z}_{\text{hard}}^* = \emptyset$ (no noise exists)

Since $\mathcal{A}$ cannot distinguish the worlds, its output must simultaneously approximate both truth sets. Any deterministic labeling misclassifies at least half the s_1 samples in at least one world. Therefore:

$$\max(\text{Error}_{\mathcal{P}_{\text{noise}}}(\mathcal{A}), \text{Error}_{\mathcal{P}_{\text{hard}}}(\mathcal{A})) \geq \frac{1}{2}$$

□

2.5 Why This Construction Is Minimal

The construction uses the **minimum number of parameters** needed to create observational equivalence between noise and difficulty:

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

Component	Why necessary	What happens if removed
2 states (s_1, s_2)	Need one ambiguous state + one clean anchor to match marginals	With 1 state, the two worlds have different $P(y)$ and $P(f_m)$ marginals, making them distinguishable
$\eta \in (0, 1/2)$	Creates the proportion of “suspicious” samples	With $\eta = 0$, both worlds are trivially identical (no noise, no difficulty)
$\varepsilon_1 \neq \varepsilon_2$	Creates inter-state expert accuracy variation	With $\varepsilon_1 = \varepsilon_2$, the s_2 anchor no longer constrains the cross-state marginal match
Expert bias in World B	Matches the noise-induced expert error pattern	Without bias, World B’s expert marginals would differ from World A’s

No simpler counterexample exists: Any construction with fewer states, fewer parameters, or simpler structure cannot simultaneously match $\mathcal{P}(y \mid x)$, $\mathcal{P}(f_m \mid x)$, and $\mathcal{P}(x)$ across two causally distinct worlds.

3 Generalization to K Classes and Arbitrary States

3.1 K -Class Construction

The binary construction extends to general K as follows:

World A (Noise): For $x \in s_1$, true label $y^* = 0$. Noise flips labels uniformly over $\mathcal{Y} \setminus \{0\}$ with probability η :

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

$$\mathcal{P}_{\text{noise}}(y = 0 \mid s_1) = 1 - \eta, \quad \mathcal{P}_{\text{noise}}(y = c \mid s_1) = \frac{\eta}{K-1}, \quad c \neq 0$$

Expert distribution: $\mathcal{P}_{\text{noise}}(f_m = 0 \mid s_1) = 1 - \varepsilon_1$, uniform over other classes.

World B (Difficulty): True label distribution matches the noise world's observation profile:

$$\mathcal{P}_{\text{hard}}(y^* = 0 \mid s_1) = 1 - \eta, \quad \mathcal{P}_{\text{hard}}(y^* = c \mid s_1) = \frac{\eta}{K-1}, \quad c \neq 0$$

No noise: $y = y^*$. Expert accuracy: given each $y^* = c$, $\mathbb{P}(f_m = c \mid s_1, y^* = c) = 1 - \varepsilon_1$, errors uniform over $K - 1$ wrong classes.

Verification: The same marginal-matching argument from Section 2.4 extends directly. The two worlds produce identical $(X, Y, \{f_m\})$ distributions but have opposite causal interpretations. $\square$

3.2 Arbitrary State Count

The K_S -state case follows by assigning each state s_i to one of two types: - **Type A** (ambiguous): follows the s_1 pattern above, with state-specific η_i and $\varepsilon_{1,i}$ - **Type B** (clean anchor): follows the s_2 pattern (no noise, constant true label)

Observational equivalence holds as long as each Type A state's parameters are matched across worlds, and Type B states provide identical constraints. This covers any $K_S \geq 1$.

4 Assumption-Identity Mapping: Which A1-A6 Breaks Which Part

Each assumption in A1-A6 breaks a specific aspect of Theorem 3’s construction. The mapping below shows that the SCX assumption set is not arbitrary but **targeted** at the specific degrees of freedom that enable unidentifiability.

4.1 Assumption-to-Construction Mapping

	What It Breaks in the Assumption Construction	Construction Element Violated
A1 (Disjoint training)	Expert independence: World B’s bias requires all experts to share the same bias pattern, which is unlikely if they trained on independent data	Experts’ prediction correlation structure differs between worlds
A2 (Con- ditional independ- ence)	The factorization $\prod_m \mathcal{P}(f_m x)$ used in verification; without it, cross-expert correlations provide an extra signal	Joint expert distribution differs
A3 (Bounded loss)	Technical (not structural); Hoeffding inequality requires boundedness	No direct construction violation
A4 (Uniform independ- ent noise)	World B’s difficulty model has input-dependent label ambiguity (η in s_1 vs 0 in s_2), violating input-independent noise	Noise rate constancy across states

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

	What It Breaks in the	Construction Element
Assumption	Construction	Violated
A5 (State homogeneity)	World B's s_1 has two subpopulations with different expert error rates ($1 - \varepsilon_1$ for $y^* = 0$ vs ε_1 for $y^* = 1$), violating within-state error uniformity	μ_s constancy within state
A6 (Balanced errors)	World B's experts are biased toward class 0, concentrating errors on the $y^* = 1 \rightarrow \hat{y} = 0$ direction, violating classwise error balance	$\max_c \mu_c(x) / (\mu_s / (K - 1))$ bounded

4.2 Breaking the Construction: Assumption Sufficiency

Each of A1, A4, A5, and A6 alone breaks the specific construction of Theorem 3. However, this does not mean any single assumption guarantees global identifiability – there may exist other unidentifiability constructions that a single assumption does not break. Hence the need for the SCX assumption set.

The **minimal assumption sets** that break all known unidentifiability constructions are:

Set 1: $\{A1, A4, A5\}$ (expert independence + uniform noise + state homogeneity)

Set 2: $\{A1, A4, A6\}$ (expert independence + uniform noise + balanced errors)

Set 3: $\{A5, A6, |\mathcal{S}| \geq 2\}$ (state homogeneity + balanced errors + multi-state structure)

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

SCX uses all six (A1-A6) to ensure robustness across all possible counterexamples and to enable the quantitative guarantees of Theorem 1.

4.3 Assumption Violation: What Degrades

Violated	Degradation Mode	Severity
A1	Expert correlations inflate FPR, Théorème 1’s exponential bound collapses	Severe
A4	State-conditional noise rates bias Δ_s estimates	Moderate
A5	Within-state error heterogeneity makes μ_s a poor summary	Moderate to severe
A6	Concentration of errors inflates FPR in minority classes	Mild to moderate
A3 (bound violation)	Hoeffding/Chernoff inapplicable; need sub-Gaussian alternatives	Technical
All six	Theorem 3’s unidentifiability takes full effect	Complete

5 Positioning Within the Literature

5.1 Measurement Error Models (Carroll et al., 2006)

The classical measurement error model posits an unobserved true variable X^* and an observed proxy $X = X^* + U$ where U is measurement error. Without

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

validation data or instrumental variables, the joint distribution of (X^*, U) is **unidentified**: infinitely many decompositions of $P(X)$ into $P(X^*) * P(U)$ are observationally equivalent.

Theorem 3 extends this result from the continuous setting to **discrete label noise with expert predictions**. The key novelty is the introduction of $\{f_m(X)\}$ as auxiliary observations. Theorem 3 shows that even with these extra observations, the identification problem persists – the measurement error (noise) and the true label distribution are not separately identifiable without structural assumptions.

Comparison:

Aspect	Classical Measurement Error	Theorem 3
Variable type	Continuous X^*	Discrete Y^* (labels)
Error model	Additive $X = X^* + U$	Label flip $Y = \text{flip}(Y^*)$
Auxiliary data	Validation data / instruments	Expert predictions $\{f_m(X)\}$
Unidentifiability	$P(X^*) * P(U)$ decomposition	Noise vs. difficulty decomposition
Resolution	Requires assumptions on U	Requires A1-A6 (minimum)

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

5.2 Label Noise Theory (Natarajan et al., 2013; Menon et al., 2015; Patrini et al., 2017)

The label noise literature establishes that the noise transition matrix T and the clean class probabilities $P(Y^* | X)$ are jointly unidentifiable from $(X, \tilde{Y})$ alone. Standard resolutions include:

- **Anchor points** (Liu & Tao, 2016): points known to belong to a specific class with certainty
- **Noise rate symmetry** (Natarajan et al., 2013): symmetric noise with known rate
- **Dual estimators** (Menon et al., 2015): use of class-probability estimators to correct for noise

Theorem 3 shows that this unidentifiability **persists and deepens** when expert predictions are added. Specifically:

1. Even with observations of $\{f_m(X)\}$, the noise vs. difficulty ambiguity remains
2. The ambiguity is **semantic**, not just statistical – two fundamentally different causal structures produce the same data
3. Resolving this ambiguity requires assumptions about the **joint structure** of experts (A1-A2), not just about the noise mechanism itself (A4)

5.3 Partial Identification (Manski, 2003; 2007)

Manski’s partial identification framework argues that when point identification is impossible, the researcher should characterize the **identified set** – the set of all parameter values consistent with the observed distribution.

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

Theorem 3 can be understood as establishing the identified set for the noise detection problem: - **Point estimation** is impossible: a single “noise flag” per sample cannot be consistently estimated - The **identified set** contains at least two distinct solutions: the “noise-driven” world and the “difficulty-driven” world - **Assumptions shrink the identified set**: each A1-A6 assumption reduces the set until point identification becomes possible (Theorem 1 regime) - **Weak features enlarge the identified set**: even with A1-A6, weak ϕ makes the SCX estimate indistinguishable from the loss baseline (Theorem 2 regime)

This positions Theorem 3 within a broader intellectual tradition: **identifiability is not binary but graded**, and the SCX framework explicitly tracks which assumptions are needed to achieve which level of identification.

5.4 Graphical Causal Models (Pearl, 2009)

From a causal perspective, Theorem 3 identifies a **Markov equivalence class** of two causal graphs:

Graph A (Noise):

$$S \rightarrow Y^* \rightarrow Y \leftarrow \text{Noise}$$

↓

$$X \rightarrow \{f_m\}$$

Graph B (Difficulty):

$$S \rightarrow Y^* (= Y)$$

↓

$$X \rightarrow \{f_m\}$$

These graphs share the same skeleton and immoralities (conditional indepen-

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

dence structure) when only $(X, Y, \{f_m\})$ are observed. They are members of the same Markov equivalence class and cannot be distinguished by any conditional independence test.

The SCX assumptions break this equivalence by imposing **additional constraints** that favor Graph A: - A1 (disjoint training): restricts the joint expert distribution - A4 (uniform independent noise): restricts the noise-induced conditional independencies - A5 (state homogeneity): restricts the within-state distribution of errors

5.5 Relationship to Dawid-Skene Identifiability

Dawid-Skene (1979) unidentifiability concerns **label permutation symmetry**: the EM likelihood has $K!$ equivalent modes corresponding to permuting class labels across annotators. This is orthogonal to Theorem 3’s concern.

	Dawid-Skene	Theorem 3
Ambiguity	Which permutation of class labels	Noise vs. difficulty
Source	Permutation symmetry of confusion matrices	Two causally distinct generative processes
Resolution	Anchor annotator, or fix one annotator’s confusion matrix	A1-A6 assumptions

Even after resolving Dawid-Skene’s label permutation problem, Theorem 3’s noise-difficulty unidentifiability persists. The two results are **orthogonal** and **cumulative**.

6 Practical Implications

6.1 Justification of the SCX Framework

1. **A1-A6 are not over-assumptions:** They are the minimal structure needed to break a fundamental identifiability failure.
2. **Logical closure:** Theorems 1-3 together cover:
 - Necessity (Theorem 3): why assumptions are needed
 - Sufficiency (Theorem 1): what is gained with them
 - Boundary (Theorem 2): when the gain vanishes
3. **Cold-start is theoretically necessary:** Theorem 3 proves that initial human review (anchor points) is not a practical compromise but a **theoretical necessity**.

6.2 What the Practitioner Must Verify

If you claim...	You must have verified...
“This sample is label noise”	A1-A6 hold, or anchor points cover the relevant state
“SCX noise detection works”	$A1-A6 + I(\phi; S) \gg 0$
“Noise and difficulty are separated”	At minimum, $A1 + A4 + A5$ (or equivalent)
“No assumptions needed”	Theorem 3 proves this is impossible

6.3 Diagnostic Checks for Assumption Violations

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

Assumption	Diagnostic	Pass threshold
A1 (disjoint)	Verify training data provenance	Known disjoint assignment
A4 (uniform noise)	χ^2 test: $P(y \neq y^* \mid x \in s)$ constant across states	$p > 0.05$
A5 (homogeneity)	KS test: within-state expert error distributions	$p > 0.05$ across states
A6 (balance)	χ^2 test: error class distributions balanced	$C_{\text{bal}} \leq 2$

6.4 Correct Interpretation of Theorem 3

What Theorem 3 does NOT say: - It does not say noise detection is impossible in practice - It does not say all methods are equally valid - It does not say SCX's assumptions are the only possible ones

What Theorem 3 DOES say: - Without explicit assumptions, noise detection lacks a well-defined ground truth - SCX's A1-A6 are exactly the structure needed - Any method that "just works" is implicitly assuming at least as much

7 Assumption Necessity Spectrum

The following diagram positions Theorem 3’s unidentifiability within a graded necessity spectrum:

No assumptions	Partial assumptions	SCX assumptions
v	v	v
Theorem 3 (full unidentifiability)	Corollaries 1-6 (partial identifiability)	Theorem 1 guarantees (point identification)
No structure	Anchor points (Cor 1) Known training (Cor 2) Uniform noise (Cor 3) State homogeneity (Cor 4) Balanced errors (Cor 5) Multi-state overlap (Cor 6)	A1 + A4 + A5 A1 + A4 + A6 A5 + A6 + S
v	v	v
Identified set = all possible processes	Identified set = partially constrained	Identified set = single point (ground truth)

References

Core Theoretical References

1. Natarajan, N., Dhillon, I. S., Ravikumar, P. K., & Tewari, A. (2013). Learning with noisy labels. *NeurIPS*.

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

2. Menon, A. K., Van Rooyen, B., Ong, C. S., & Williamson, R. C. (2015). Learning from corrupted binary labels via class-probability estimation. *ICML*.
3. Patrini, G., Rozza, A., Menon, A. K., Nock, R., & Qu, L. (2017). Making deep neural networks robust to label noise: A loss correction approach. *CVPR*.

Measurement Error Models

4. Carroll, R. J., Ruppert, D., Stefanski, L. A., & Crainiceanu, C. M. (2006). *Measurement Error in Nonlinear Models: A Modern Perspective* (2nd ed.). Chapman and Hall/CRC.
5. Fuller, W. A. (1987). *Measurement Error Models*. Wiley.

Partial Identification

6. Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer.
7. Manski, C. F. (2007). *Identification for Prediction and Decision*. Harvard University Press.

Graphical Causal Models

8. Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

Finite Mixture Models

9. Teicher, H. (1963). Identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 34(4), 1265-1269.
10. McLachlan, G. J., & Peel, D. (2000). *Finite Mixture Models*. Wiley.

Multi-Annotator Models

11. Dawid, A. P., & Skene, A. M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. *JRSS Series C*, 28(1), 20-28.
12. Northcutt, C. G., Jiang, L., & Chuang, I. L. (2021). Confident learning: Estimating uncertainty in dataset labels. *JAIR*, 70, 1373-1411.

Related SCX Documents

13. SCX Theorem 1 (Polished): Multi-Expert Consistency Guarantees. `./01_noise_detection_polished.md`.
14. SCX Theorem 2 (Polished): Weak Feature Failure Lower Bound. `./02_weak_feature_polished.md`.
15. Unified Notation and Dependencies. `./00_notation_and_dependencies.md`.
16. SCX Framework Definitions. `../definitions/01_state_conditioned_risk.md`.

Revision notes (2026-06-27): 1. **Generalized proof:** Extracted minimality argument; K -class construction added; arbitrary state count addressed. 2. **Assumption-identity mapping:** New Section 4 mapping each A1-A6 assumption to the construction element it breaks. 3. **Literature connections:** Added Sections 5.1 (Carroll measurement error), 5.2

THEOREM 3 (POLISHED): UNIDENTIFIABILITY OF NOISE VS. LEARNABLE DIFFICULTY

(Natarajan label noise), 5.3 (Manski partial identification), 5.4 (Pearl causality). 4. **Practical implications:** New Section 6 with diagnostic checks and mistaken-interpretation guardrails. 5. **Assumption necessity spectrum:** New Section 7 positioning Theorem 3 within a graded identifiability framework.